Smart Helmet using IOT for Alcohol Sensing and Intelligent Accident Detection

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Abstract—**The primary focus of the paper revolves around leveraging advanced technology to mitigate the risks associated with drunk driving among motorcyclists while simultaneously enhancing safety through swift accident detection and response mechanisms. By integrating IoT capabilities into the helmet, it becomes a proactive tool in preventing potentially fatal incidents on the roads. One pivotal aspect of this innovation is its ability to detect alcohol consumption levels among riders. Through specialized sensors embedded within the helmet, it can analyze the wearer's breath or perspiration for traces of alcohol, thereby acting as a pre-emptive measure against intoxicated riding. This feature serves as a crucial deterrent, discouraging individuals from engaging in irresponsible behavior that jeopardizes their safety and that of others on the road.**

Keywords—GPS-global positioning system, MQ3, GSM, ADXL335.

I. INTRODUCTION

With the escalating rate of accidents, governments worldwide are implementing stringent laws and regulations to curb these incidents. Accidents, defined as unforeseen events leading to injury or fatality, are particularly prevalent among two-wheeler riders. The use of helmets and refraining from alcohol consumption while riding significantly mitigates these risks. This paper conducts a survey on smart helmet systems designed for accident avoidance, exploring various techniques. Additionally, it delves into IoT technology, which has gained prominence in recent times. Through literature review, it is found that systems employing microcontrollers, RF transmitters, and other sensors are cost-effective. However, those utilizing Raspberry Pi modules, Pi cameras, pressure sensors, and GPS systems, integrating image processing algorithms, emerge as the most efficient. Incorporating image processing enables the detection of helmet usage by the rider. Smart helmet systems contribute to enhancing safety and security for two-wheeler riders.

This paper aims to reduce the road accidents caused by drunken drivers, which is a major cause of fatalities worldwide. Moreover, the smart helmet's intelligent accident detection system represents a groundbreaking advancement in road safety technology. Equipped with sophisticated sensors and algorithms, it can swiftly discern sudden impacts or falls experienced by the rider. Upon detection of such events, the helmet initiates an automated alerting process, promptly notifying emergency services or designated contacts. This rapid response mechanism ensures that necessary assistance is summoned without delay, significantly enhancing the likelihood of timely medical intervention and minimizing the severity of potential injuries. In essence, the overarching objective of the proposed method is to foster a safer and more responsible riding environment by seamlessly integrating cutting-edge technology into the realm of motorcycle safety. By proactively addressing the dangers of drunk driving and swiftly responding to accidents, this innovative solution strives to save lives, reduce injuries, and promote greater awareness of road safety practices among riders.

The proposed method is an overview of the alcohol detection and the engine turning off system. This method also has the advantage of the accident detection system with location sharing and a message intimation system, And also this idea has increased number of benefits in Personal safety: The system also ensures the safety of the helmet wearer. By preventing the engine from starting if the alcohol level is above the set limit[threshold value - 500], it helps protect the individual from potential harm that may arise due to impaired judgment and decreased reaction time while driving.

II. LITERATURE SURVEY

1. **K.M. Mehta et al.**: They proposed a system focused on worker safety, particularly fall detection in working areas. Their system comprises a wearable device with sensors and electronic elements, along with a cell phone. Communication between these components is facilitated by a GSM module. Continuous monitoring of worker health and safety

is ensured, with the system promptly detecting falls and alerting designated personnel for medical attention.

2. **Divyasudha N et al.**: Their system involves a microcontroller, position sensor, alcohol sensor, piezoelectric sensor, RF transmitter, IoT modem, GPS receiver, power supply, and solar panel. It aims to prevent accidents and monitor alcohol consumption among riders. The system checks if the rider is wearing a helmet and whether they have consumed alcohol. Failure to comply results in the bike not starting, signaled by a beep sound. In the event of an accident, predefined contacts and the police station are notified via IoT modem.
3. **Manish Uniyal et al.**: Their proposed system consists of a helmet unit and a two-wheeler unit. The helmet unit communicates its position data to the two-wheeler section using an RF receiver. Various sensors, including an accelerometer, Hall-effect sensor, and GPS module, are placed on the two-wheeler to collect data. This data is sent to a server if there is an internet connection, allowing real-time monitoring of vehicle speed and helmet usage. Parents can monitor whether their child is wearing a helmet.
4. **Shoeb Ahmed Shabbeer et al.**: They proposed a smart helmet method for accident detection and reporting. Their system utilizes a microcontroller interfaced with an accelerometer and GSM module. Accidents are detected based on acceleration exceeding a threshold, with notifications sent to emergency contacts via GPS module and cloud infrastructure. The system achieved high accuracy in accident identification and coordinate transmission.
5. **P. Roja et al.**: Their system comprises six units, including a remover sensor, IR sensor, air quality sensor, Arduino Uno microcontroller, GPRS, and GSM. Designed for mining environments, the helmet alerts workers to harmful gases and transmits data to the server if the helmet is removed. IoT technology facilitates data transmission in this system.

These systems demonstrate diverse approaches to enhancing safety and monitoring in various contexts, from worker safety in industrial environments to accident prevention and detection in transportation settings. Each system leverages different sensors, communication technologies, and monitoring mechanisms tailored to specific application scenarios.

III. RESEARCH METHODOLOGY

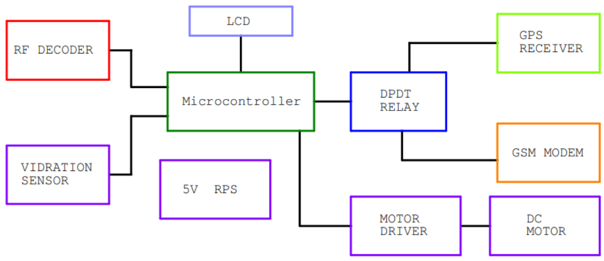

Fig 1 - Block diagram OF THE PROPOSED METHOD.

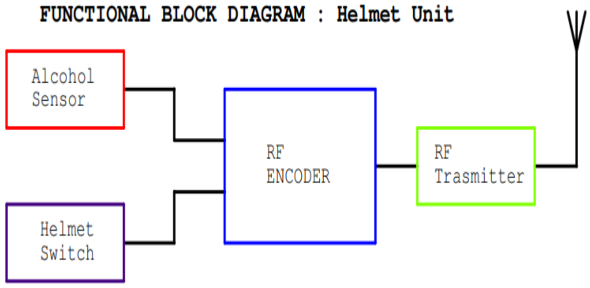

Fig 2 - Block diagram 2

IV. PROPOSED SYSTEM SPECIFICATION

a) Alcohol Sensor :The presence of alcohol in the breath is detected by an alcohol sensor. The power supply will only be 150 mV to 5V.MQ-3 is a well known type of alcohol sensor that can be used with a breath analyzer. The sensing range of MQ-3 is 0.04 mg/L to 4 mg/L.

b) GPS NEO 6m: GPS is used to navigate a location based on latitude and longitude and is used for continuous tracking of a vehicle.

c) GSM 800L Module: Global System for Mobile Communication is the abbreviated form. It is a module that is built for sending SMS wirelessly .

*d) Accelerometer or Vibration Sensor[ADXL 335] :*A vibration sensor is a device that detects sudden vibration in a system, Operating voltage is DC 5-15V. Frequency range from 0.2 up to 2500 Hz with a sensitivity of 100 mV/g.

e) LM2596 :The LM2596 is a popular voltage regulator integrated circuit (IC) that is widely used in electronic circuits to step down voltage. It is commonly used in power supply applications to provide a stable and adjustable output voltage from a higher input voltage. The LM2596 is available in various packages, and there are also pre-built modules or boards based on this IC for ease of use.

V. PROPOSED WORK

Our innovative system utilizes a specialized alcohol sensor (MQ3) to accurately detect the presence of alcohol in a person's breath, thus addressing the critical issue of drunk driving and preventing potential accidents. Integrated with an Arduino board, this sensor communicates its findings, triggering a series of automated responses aimed at ensuring safety on the road.

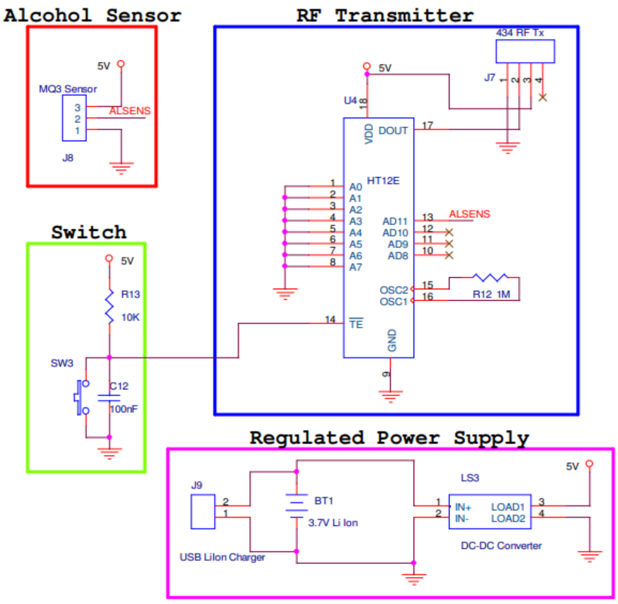

Fig 3 - Circuit Diagram of Helmet Unit

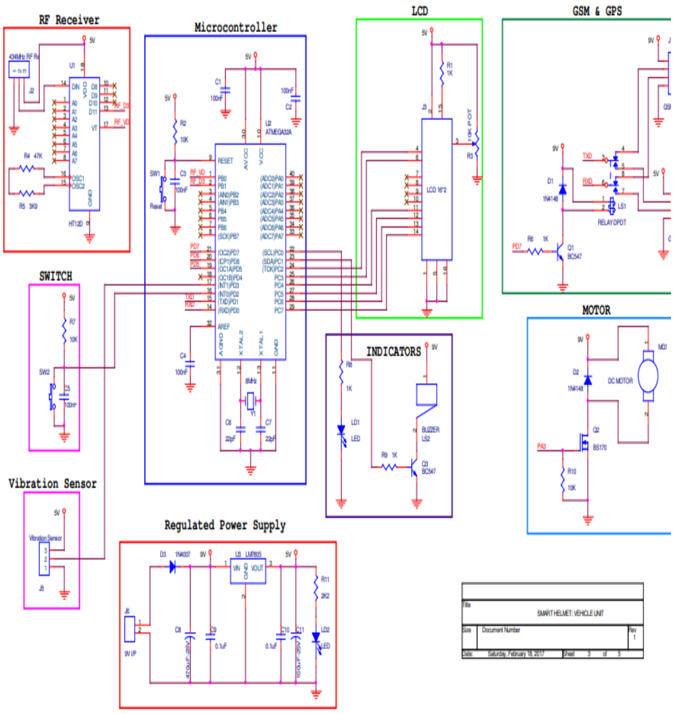

Fig 4 - Circuit Diagram of Vehicle Unit

Upon detecting alcohol, the Arduino takes immediate control, activating various components such as the vehicle motor, an LCD screen, and a buzzer. A warning sound alerts the user, while the LCD displays a clear message indicating alcohol detection. Crucially, the vehicle is rendered inoperable, preventing any further risk of driving under the influence.

Furthermore, our system goes beyond alcohol detection to offer comprehensive accident prevention and response capabilities. In the event of an accident, sensors swiftly detect the impact, prompting the system to initiate a multi-tiered emergency protocol. Notifying nearby emergency services immediately, it ensures prompt assistance is dispatched to the scene. Moreover, our system prioritizes communication with the user's designated family contacts, sending them notifications and making calls to apprise them of the situation. By sharing the precise location of the incident obtained from GPS, loved ones are kept informed and empowered to provide support as needed.

In essence, the proposed work serves as a reliable safety companion, constantly vigilant and proactive in safeguarding individuals on the road. With its integrated alcohol sensing, accident detection, and real-time communication capabilities, our system offers peace of mind and enhanced protection for riders and their loved ones, reinforcing responsible and safe driving practices.

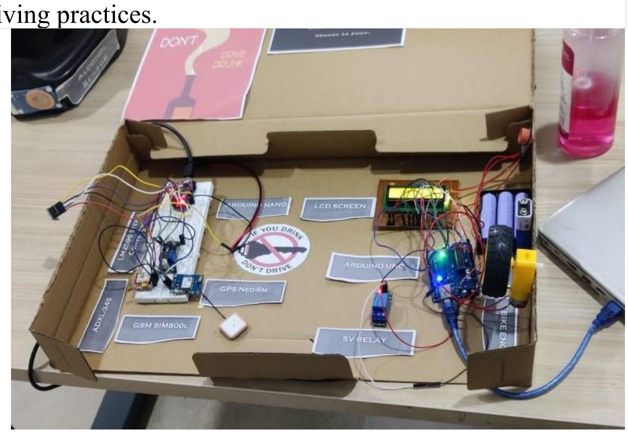

Fig 5 - Built prototype

VI. CONCLUSION

The core objective of the system is to bolster safety measures by swiftly identifying accidents and promptly notifying the user or emergency services. Additionally, the system plays a pivotal role in curbing the dangers of impaired driving by detecting alcohol levels and effectively immobilizing the vehicle's engine. By preventing individuals under the influence from operating vehicles, the system significantly enhances overall road safety, thereby fostering a safer and more responsible environment for all road users.

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